Threat Hunting Report

Introduction

This practical exercise focused on performing threat hunting using a hypothesis-driven approach. The objective was to detect potential unauthorized privilege escalation by analyzing logs in Elastic Security, validating indicators using AlienVault OTX, and confirming endpoint activity through Velociraptor.

The investigation was aligned with **MITRE ATT&CK technique T1078 (Valid Accounts)**, which involves misuse of legitimate credentials for unauthorized access.

Tools Used

Tool	Purpose
Elastic Security	Log analysis and event monitoring
Velociraptor	Endpoint investigation and process analysis
AlienVault OTX	Threat intelligence and IOC validation

Threat Hunting Hypothesis

Unauthorized privilege escalation is occurring using valid domain accounts.

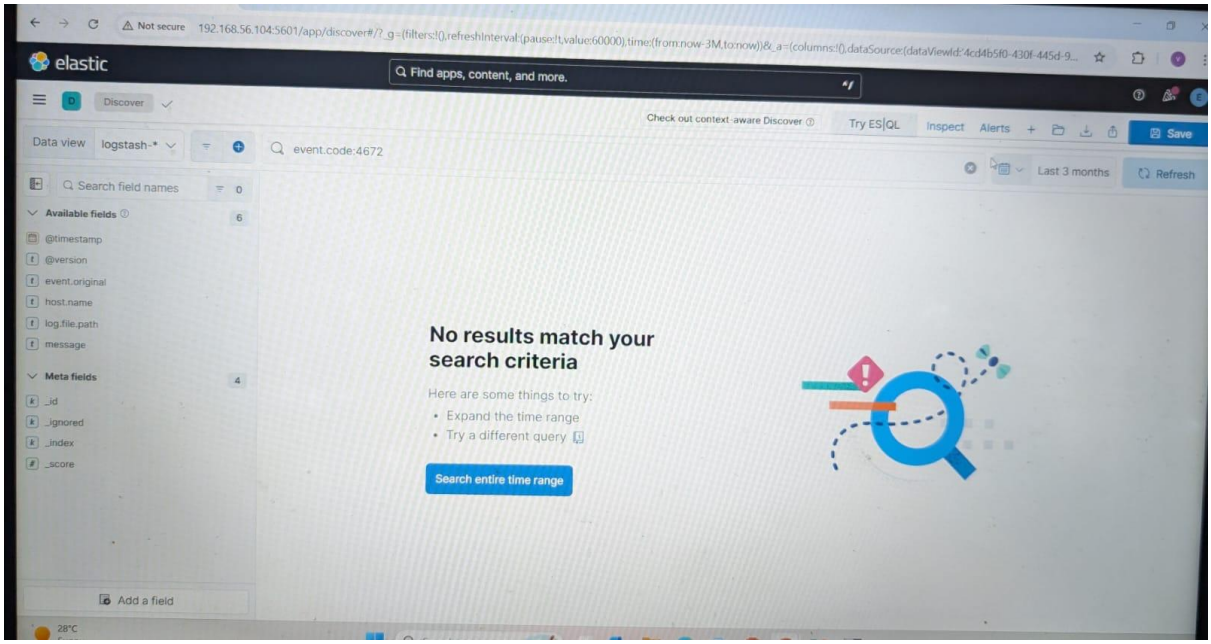
This hypothesis is based on the assumption that attackers may use legitimate credentials to gain elevated privileges.

Step 1: Log Analysis in Elastic Security

The first step involved querying logs in Elastic Security to identify privilege escalation events.

Query Used

```
event.code:4672
```



Observation

- No results were returned for Event ID 4672
- This indicates:
  - No recorded privilege escalation events
  - OR logs not available / not ingested properly

Conclusion

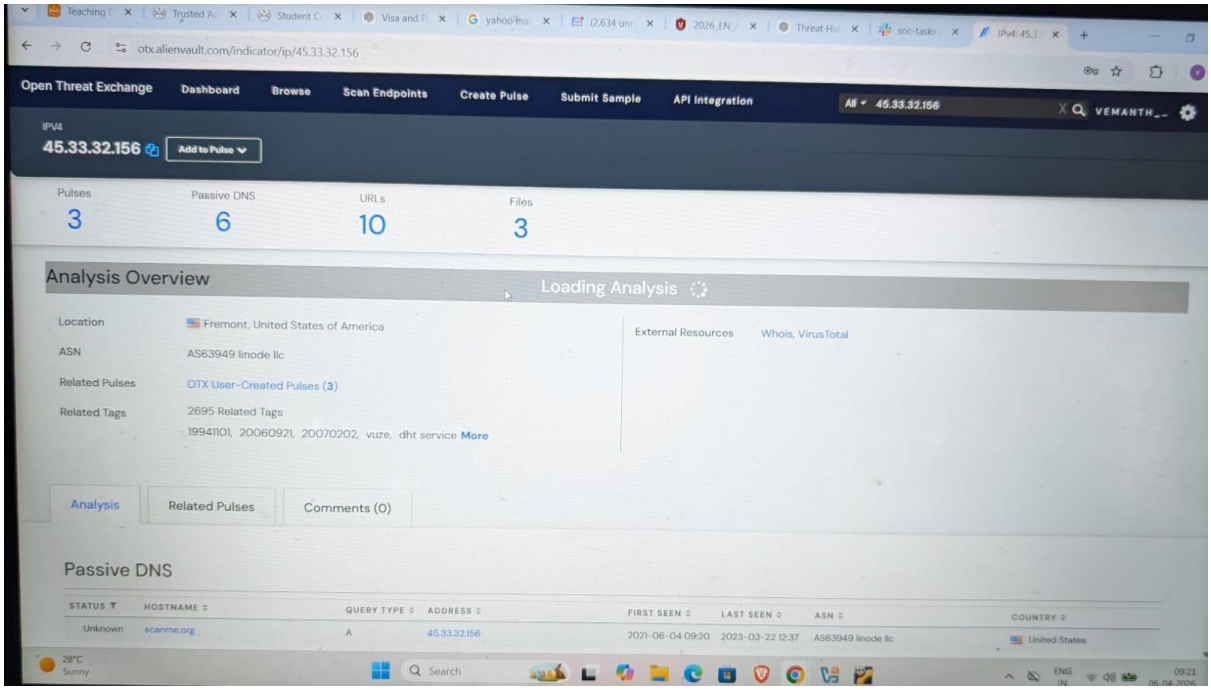
Although no direct evidence of privilege escalation was found in Elastic, further investigation was required using other tools.

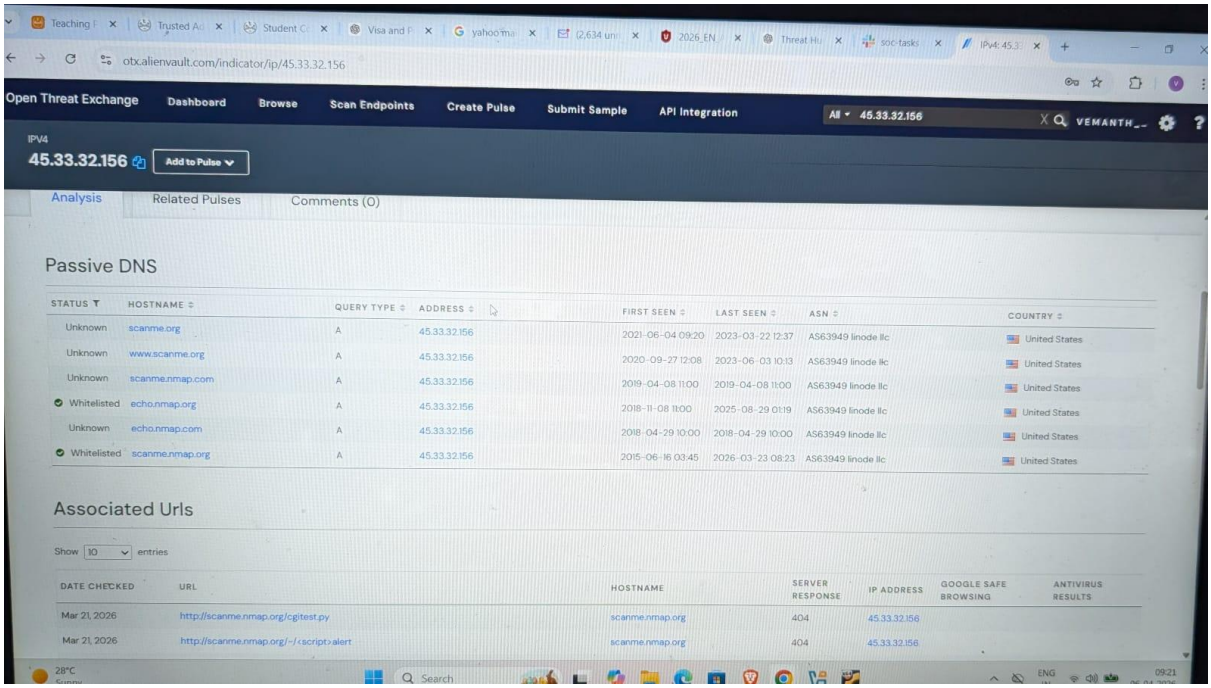
Step 2: Threat Intelligence Hunt (AlienVault OTX)

The next step involved validating suspicious IP addresses using AlienVault OTX.

IOC Searched

```
45.33.32.156
```





Findings

IP associated with multiple domains:

- `scanme.org`
- `scanme.nmap.com`

- Hosted in:
  - United States (Linode ASN)
- Passive DNS shows repeated activity

Analysis

- The IP is linked to **Nmap scanning infrastructure**
- It is not strictly malicious but can indicate:
  - Network scanning activity
  - Reconnaissance behavior

Step 3: Endpoint Investigation using Velociraptor

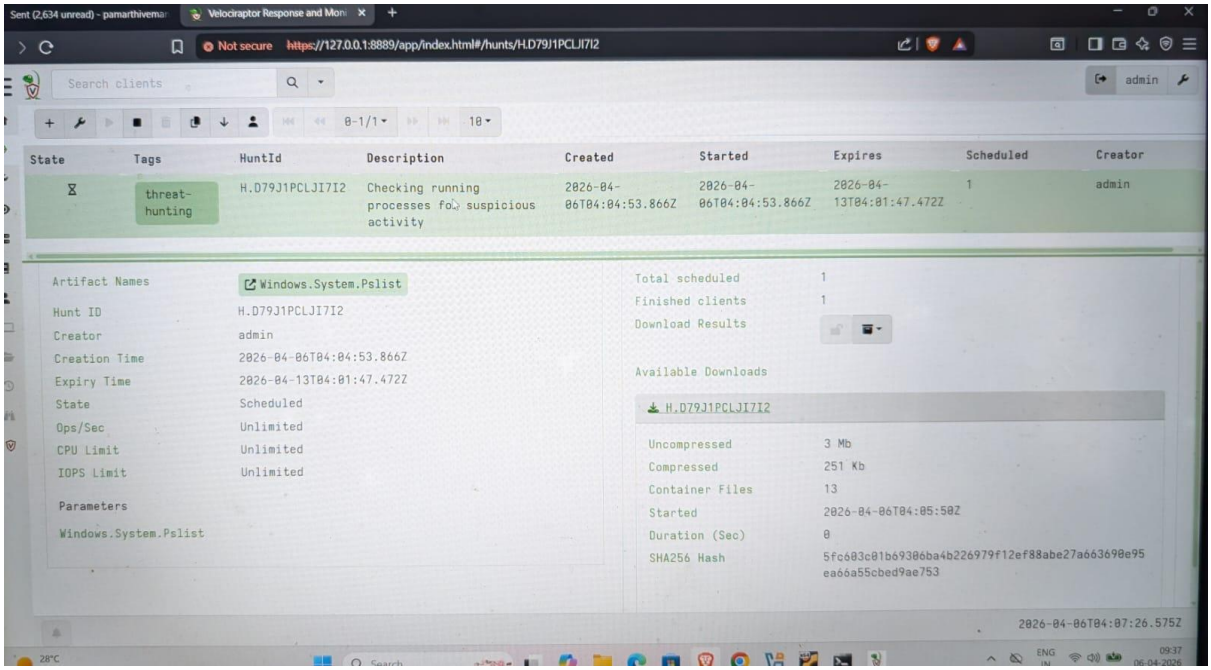
A hunt was created using Velociraptor to analyze running processes.

Artifact Used

`Windows.System.Pslist`

Hunt Details

- Tag: threat-hunting
- Description: Checking running processes for suspicious activity
- Status: Completed successfully



Findings

- Process data was successfully collected
- No immediate critical anomalies visible in summary
- Full results required further inspection (downloaded artifact)

Analysis

- System processes appear normal
- No strong evidence of malicious execution
- However, process monitoring is essential for detecting hidden threats

Correlation of Findings

Source	Observation
Elastic	No Event ID 4672 detected
OTX	IP linked to scanning activity
Velociraptor	Processes appear normal

Threat Analysis

The investigation did not reveal direct evidence of privilege escalation. However:

- External IP (45.33.32.156) indicates **reconnaissance activity**
- Lack of logs in Elastic highlights **visibility gap**
- Endpoint processes appear normal, suggesting no active compromise

MITRE ATT&CK Mapping

Technique ID	Technique Name	Description
T1078	Valid Accounts	Use of legitimate credentials for unauthorized access

Hunting Report

The threat hunting investigation aimed to detect unauthorized privilege escalation using Elastic Security, Velociraptor, and AlienVault OTX. No Event ID 4672 logs were found in Elastic, indicating no direct evidence of privilege escalation or insufficient log visibility. Threat intelligence analysis of IP address 45.33.32.156 revealed associations with known scanning infrastructure, suggesting reconnaissance activity. Velociraptor analysis of running processes did not identify any suspicious behavior, indicating no active compromise at the endpoint level. Although no confirmed attack was detected, the presence of scanning activity highlights potential external threats. The findings emphasize the importance of continuous monitoring and improved log collection for effective threat detection.

Conclusion

This practical exercise demonstrated a complete threat hunting workflow using multiple tools. While no active compromise was detected, the investigation revealed potential reconnaissance activity and highlighted limitations in log visibility.

The exercise reinforces the importance of combining log analysis, threat intelligence, and endpoint investigation to detect advanced threats effectively.